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Denver Regional Transportation District

Denver EMU Eagle P3 Door System

Project # 7230

Door System

Software Applications Configuration Specifications (SACS)

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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Document No.	**201111576**	Revision	**2**
		Date (yyyy-mm-dd)	**2011-11-14**

Revision List

Rev.	Date	By	ECD	Description of changes
2	2011-11-14	James Kim	-	Official release
1.2	2011-11-14	James Kim	-	Implemented comments from internal inspection I-1892
1.1	2011-10-30	José Trejo & James Kim	-	**Change according to the Rotem's comment by email on 2011-10-18 "RE: Denver Generic Document":** • In Figure 1-1, added SPI SDD (201111659). **Changes according to the internal review:** • Throughout the document, IBM RequisitePro tag is added. • In Table 3-4, swapped the "Valid Values" of "Crew Switch Closing Time" and "Crew Switch Opening Time". • In Table 3-6: Event List: o Description for Event 22 is changed. o Event 53 changed to have the S/R type for each L1 and R1. o Event 66 changed to have the S/R type for each L1 and R1. o Event 70 changed to have the S/R type for each L1 and R1. o Event 74 changed to have the S/R type for each L1 and R1. • In Section 3.1.3.1 "Door operation movement function", added VAP5.1.1.2.2 tag and sentence. • In Section 3.1.3.1, added VAP5.1.7.3 Valid Passenger Door Open Command: Changed to "Valid Passenger Door Open Pushbutton Command" • In Section 3.1.3.1, added sentence that Crew Switch does not affect the Visual Alarm nor Audible Alarm.
1	2011-09-20	James Kim	-	Official release
0.1	2011-09-19	James Kim	-	Implementing comments from internal inspection I-1831
0	2011-09-14	James Kim	-	Original issue

Table of Contents

REVISION LIST **2**

TABLE OF CONTENTS **3**

LIST OF FIGURES **4**

LIST OF TABLES **4**

1 INTRODUCTION **5**

1.5 Overview 8

2 BOOT LOADER **10**

2.1 Software Requirement Specification Document 10
2.2 Software Design Description Document 10
2.2.1 PTU Login String 10
2.2.2 Processor Information 10
2.2.3 Operation System Information 10
2.2.4 Memory Address 11
2.2.5 Code Information Address 11
2.2.6 Memory Layout 12
2.3 Software Integration Test Plan Document 12

3 DCU APPLICATION **13**

3.1 Software Requirement Specification Document 13
3.1.1 Denver Input and Output Signal Mapping 13
3.1.2 External Interfaces 15
3.1.3 Specific Functions 16
3.1.4 DCU Event List 19
3.2 Software Design Description Document 27
3.2.1 Serial Driver Module SDD Document 27
3.2.2 Watchdog Module SDD Document 27
3.2.3 Door Operator Module SDD Document 27
3.2.4 Fault Logger Module SDD Document 27
3.2.5 Proportional Integral Derivative Module SDD Document 27
3.2.6 Portable Test Application Module SDD Document 27
3.2.7 Real-Time Clock Module SDD Document 27
3.2.8 Real Time Data Monitoring Module SDD Document 28
3.2.9 Neuron SPI Module SDD Document 28
3.2.10 Self-Test Module SDD Document 28

4 PTU APPLICATION **29**

4.1.1 Software Requirement Specification Document 29
4.1.2 Software Design Description Document 29

List of Figures

Figure 1-1: Hierarchy of the Document 9

List of Tables

Table 2-1: Memory Address 11
Table 2-2: Code Information Address 11
Table 2-3: Memory Mapping on DCU 12
Table 2-4: Static RAM Memory Mapping 12
Table 3-1: Signal Mapping with Denver Project 13
Table 3-2: List of the Input from the Door Control Panel 15
Table 3-3: List of the Input and Output from the PDOP 15
Table 3-4: PTU Configurable Door Parameter 17
Table 3-5: Snapshot Specification in Denver Project 18
Table 3-6: Event List 19
Table 3-7: Event Detected by Door Operator Module 27
Table 3-8: Event Detected by the Portable Test Application Module 27
Table 3-9: Event Detected by Self-Test Module 28

1 INTRODUCTION

This document is the Software Applications Configuration Specifications (SACS) for the Door System softwares developed by Vapor Stone Rail Systems.
This document has to be considered in conjunction with the generic software documentation to fullfill the project Regional Transportation District (RTD) Denver EMU's software requirement.

1.1 Purpose

The purpose of this document is to describe the specific elements of the project Denver EMU Eagle P3 Door System in addition to the generic software documentation to cover all clients' requirements.

1.2 Scope

The scope of this document covers the Boot Loader, the DCU Application, and the PTU Application software.

1.3 Definitions, acronyms, and abbreviations

Definitions and acronyms listed below establish meaning in the context of this system requirements specifications document.

1.3.1 Definitions

1.3.1.1 Denver Project Definition

Boot Loader: Startup software verifies the memory and integrity of the application before jumping to the main application.

Door Control Unit: A processor based electronic unit running software that controls the door, provides monitoring functions, and interacts with the user via a PTU Application and a network.

1.3.1.2 Generic Documentation Definition

Door Control Module: Door Control Module (DCM) Application generic documents (SRS and SDD) use this term. It is a specific processor based electronic module running a Door Control Module Application. This application controls the door motor. It is called the Door Control Unit (DCU) Application in the Denver project.

Door Module System: Boot Loader Application generic documents (SRS, SDD, and SITP) use this term. It generically identifies a processor based electronic module that runs any software within any controller.

HW_COM: Communication interface used for communication between Portable Test Application and Door Module System application, that can be USB, serial or Ethernet**.**

Portable Test Application (PTA): Portable Test Application (PTA) is a term used in the generic documentation, and it is called the Portable Test Unit (PTU) Application in the Denver project.

1.3.1.3 Event Level Definition

Event: Term used to globally identify an Info, an Exception or a Failure.

Info: Info is an occurrence of an Informative condition monitored by the system, which is neither an exception nor a failure.

Exception: An exception is an occurrence of a condition that is outside of the specified parameters and affecting an outside Line Replaceable Unit (LRU).

Failure: A failure is an occurrence of a condition that is outside of the specified parameters and affecting this LRU. A failure normally requires maintenance or replacement to restore affected unit to its normal operation conditions.

1.3.2 Acronyms

Acronym	Name
C/B	Circuit Breaker
CL	Car Line
CNC	Car Network Controller
CPLD	Complex Programmable Logic Device
CPU	Central Process Unit
CRC	Cyclic Redundancy Check
CS	Crew Switch
DCL	Door Control Logic
DCM	Door Control Module
DCML	Door Cut-out Mechanical Lock
DCP	Door Control Panel
DCU	Door Control Unit
DMI	Door Movement Information
DMS	Door Module System
DO	Door Operator
DOP	Door Operator
DSR	Door Summary Relay
EDF	Event Description File
EMS	Emergency Handle Switch
EMU	Electric Multiple Units
I/O	Input / Output
ICD	Interface Control Document
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers, Inc.
IRAM	Internal Random Access Memory
ITPS	Integration Test Procedure Specification
LED	Light-Emitting Diode
LRU	Line Replaceable Unit
MDC	Master Door Controller
MDS	Monitoring and Diagnostic System
MLS	Mechanical Lock Signal
NGA	Neuron Gateway Application
NVRAM	Non-Volatile Random Access Memory
OPDF	Operating Parameters Description File
OS	Operation System
PDOP	Passenger Door Open Pushbutton
PID	Proportional Integral Derivative
PSS	Panel Sensing Switch
PTA	Portable Test Application
PWM	Pulse Width Modulation

Acronym	Name
RTD	Regional Transportation District
RTDM	Real-Time Data Monitoring
S/R	Set / Reset
SACS	Software Applications Configuration Specification
SCI	Software Configuration Item
SDD	Software Design Description
SFD	System Functional Description
SITP	Software Integration Test Plan
SLR	Signal Light Relay
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
SRS	System Requirements Specifications
SSL	Sensing Switch Left
SSR	Sensing Switch Right
TL	Train Line
TS	Technical Specifications
UART	Universal Asynchronous Receiver Transmitter
XRAM	On-Chip Extension Random Access Memory

1.4 References

This section lists the number, title, revision, date, and source of all documents referenced in this document.

Standards

IEEE Std 1558-2004 IEEE recommended practice for software documentation for rail equipment and systems, 2005-03-23, IEEE, www.ieee.org

Vapor Stone Rail Systems documents

201111185 System Functional Description – Door System, Rev. 4

201111270 Software Requirement Specification - Generic Door Control Module Application, Rev. 1

201111271 Software Requirements Specification - Generic Boot Loader Application, Rev. 1

201111274 Software Requirements Specification - Generic Portable Test Application, Rev. 1

201111295 Software Design Description - Generic Portable Test Application, Rev. 1

201111450 Master Software Design Description – Door Control Unit Application, Rev. 1

201111453 Software Integrated Test Plan - Door Control Unit Application, Rev. 1

201111456 Software Design Description - Generic Boot Loader Application, Rev. 1

201111464 Software Integrated Test Plan - Portable Test Unit Application, Rev. 1

201111544 Software Design Description – Generic Watchdog Module, Rev. 1

201111554 Software Design Description – Generic Proportional Integral Derivative Module, Rev. 1

201111568 Software Design Description – Generic Door Operator Module, Rev. 1

201111580	Software Integrated Test Plan - Generic Boot Loader Application, Rev. 1
201111581	Software Design Description – Generic Real Time Clock Module, Rev. 1
201111582	Software Design Description – Generic Serial Driver Module, Rev. 1
201111583	Software Design Description – Generic Fault Logger Module, Rev. 1
201111584	Software Design Description – Generic Self-Test Module, Rev. 1
201111585	Software Design Description – Generic Real Time Data Monitoring Module, Rev. 1
201111586	Software Design Description – Generic Portable Test Application Module, Rev. 1
201111628	Software Design Description – Input / Output Module, Rev. 1
201111629	Interface Control Document - Serial Peripheral Interface Protocol, Rev. 1
201111659	Software Design Description – Generic Neuron SPI Module, Rev. 1

1.5 Overview

As seen in the Figure 1-1, this document covers three SCI, and it is composed of four sections with following information:

- Section 1, Introduction states the purpose, identifies the scope, gives definitions, abbreviations and an overview of this document.
- Section 2, Boot Loader provides specific aspects of following documents:
    - Software Requirement Specification Document
    - Software Design Description Document
    - Software Integration Test Plan Document
- Section 3, VAP5 DCU Application provides specific aspects of following documents:
    - Software Requirement Specification Document
    - Software Design Description Document
- Section 4, PTU Application provides specific aspects of following documents:
    - Software Requirement Specification Document
    - Software Design Description Document

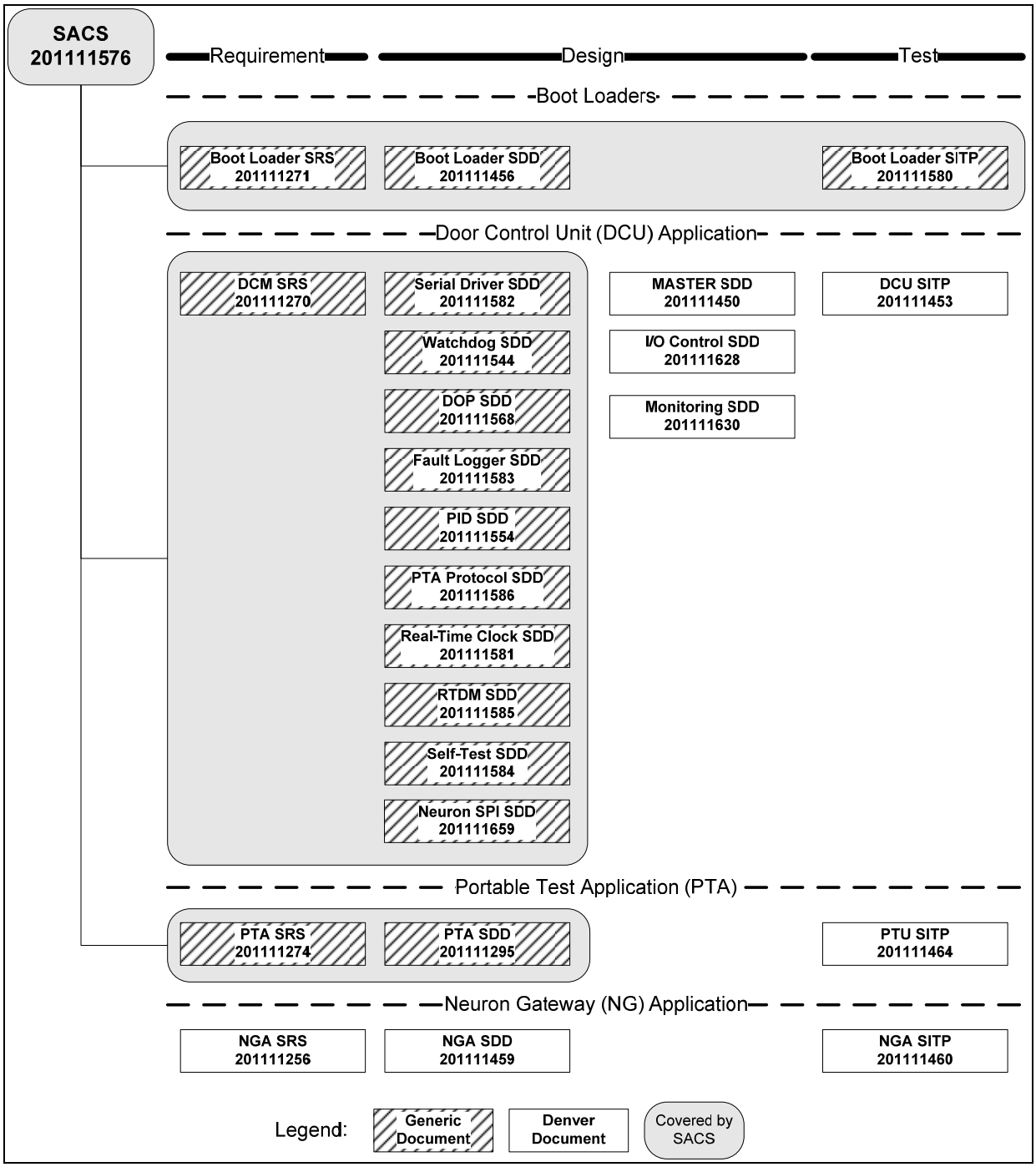

Figure 1-1: Hierarchy of the Document

2 BOOT LOADER

2.1 Software Requirement Specification Document

The Generic Boot Loader SRS #201111271 has a complete requirement specification for this project.

2.2 Software Design Description Document

The Generic Boot Loader SDD #201111456 requires a specific design description, and following sub-sections describes them.

2.2.1 PTU Login String

There are two login strings: the login request string from the PTU Application and the Login response string from the Boot Loader.

- The login request string is “Denver, PTU Door Module System, VSRS”
- The login response string is “Denver, Door Module System, VSRS’

2.2.2 Processor Information

The processor used for this project is ST10F276.

2.2.3 Operation System Information

The Operation System used for this project is:

- OS Company Name: Keil Elektronik
- OS Name: Keil ARTX-166 Real-Time OS
- OS Version: Version 2.02a

2.2.4 Memory Address

The memory address used for this project is:

Table 2-1: Memory Address

Memory	Address	Size (byte)
XRAM	0xE000~0xE7FF	2,048
IRAM	0xF600~0xFDFF	2,048
IFlash	Block 1: 0x00’0000~0x00’1FFF	8,192
	Block 2: 0x00’2000~0x00’3FFF	8,192
	Block 3: 0x00’4000~0x00’5FFF	8,192
	Block 4: 0x00’6000~0x00’7FFF	8,192
	Block 5: 0x01’8000~0x01’FFFF	32,768
	Block 6: 0x02’0000~0x02’FFFF	65,536
	Block 7: 0x03’0000~0x03’FFFF	65,536
	Block 8: 0x04’0000~0x04’FFFF	65,536
	Block 9: 0x05’0000~0x05’FFFF	65,536
	Block 10: 0x06’0000~0x06’FFFF	65,536
	Block 11: 0x07’0000~0x07’FFFF	65,536
	Block 12: 0x08’0000~0x08’FFFF	65,536
XFlash	Block 1: 0x09’0000~0x9’FFFF	65,536
	Block 2: 0x0A’0000~0xA’FFFF	65,536
	Block 3: 0x0B’0000~0xB’FFFF	65,536
	Block 4: 0x0C’0000~0xC’FFFF	65,536
	Block 5: 0x0D’0000~0xD’FFFF	65,536
SRAM	0x0F’0000~0x10’0000	65,536
NVRAM	0x12’0000~0x12’7FFF	32,768

2.2.5 Code Information Address

The address of code information for the Boot Loader and the Application are:

Table 2-2: Code Information Address

Code Information	Address	Size (byte)
Boot Loader	0x00200	48
Application	0x20200	48

2.2.6 Memory Layout

The memory layout is defined as following:

Table 2-3: Memory Mapping on DCU

Start Address	End Address	Size	Description
0x000000	0x007FFF	32 KB	Boot Loader Flash Code Part 1
0x008000	0x00DFFF	24 KB	Unused Flash Memory
0x00E000	0x00E7FF	2 KB	Internal XRAM
0x00E800	0x00F5FF	3.5 KB	Special Function Register
0x00F600	0x00FDFF	2 KB	Internal IRAM
0x00FE00	0x00FFFF	512 bytes	Special Function Register
0x018000	0x01FFFF	32 KB	Boot Loader Flash Code Part 2
0x020000	0x07FFFF	448 KB	Application Code
0x080000	0x0DFFFF	384 KB	Data flash (for expansion)
0x0F0000	0x0FFFFF	64 KB	Static RAM
0x120000	0x127FFF	32 KB	NVRAM
0x128000	0xFFFFFF	15200 KB	Free

Table 2-4: Static RAM Memory Mapping

Start Address	End Address	Class	Size	Description
0x0F0000	0x0F01FF	FCODE	512 bytes	Interrupt Vector Table
0x0F0220	0x0F0221	ND_CPU	2 bytes	CPU Usage in %
0x0F0222	0x0F0225	***Reserved***	4 bytes	RTC
0x0F0226	0x0F07FF	NDATA	1498 bytes	General Register
0x0F0A9E	0x0F0FFF	NRXBUF	1378 bytes	Communication Buffer
0x0F1000	0x0F11FF	N_CSTK	256 bytes	C User Stack
0x0F1200	0x0F28FF	***Reserved***	5.75 KB	
0x0F2900	0x0F3FFF		5.75 KB	***Available***
0x0F4000	0x0F7FFF	HOSSTK	16.00 KB	OS Stack
0x0F8000	0x0FE3FF	***Reserved***	25.00 KB	*Event Faults Log(Application)*
0x0FD400	0x11FFFF		75.00 KB	***Available***
0x127F00	0x127F1F	FD_MEM	32 bytes	*Data backup of memory check*

2.3 Software Integration Test Plan Document

The Generic Boot Loader ITPS #201111580 has a complete test procedure for this project.

3 VAP5 DCU APPLICATION

3.1 Software Requirement Specification Document

The Generic DCM Application SRS #201111270 requires a specific requirement specification, and following sub-sections describes them.

3.1.1 Denver Input and Output Signal Mapping

Table 3-1 maps the name of the signals used in the Denver project with the ones from the Generic DCM Application SRS.

Table 3-1: Signal Mapping with Denver Project

Denver Door System Signal	Mnemonic in SRS
DCU External Signal	
Zero Speed (+) Car Line	Input.NoMotion.TL.Pwr.ISO
Zero Speed (-) Car Line	Input.NoMotion.TL.Rtn.ISO
Door Open (+) Train Line	Input.DoorOpenTL.ISO
Door Open (-) Train Line	Input.DoorUnlockTL.ISO
Door Close Train Line	Input.DoorCloseTL.ISO
Passenger Door Enable (+) TL	-
Passenger Door Enable (-) TL	-
Crew Switches Open	Input.Crew.Out.Open.Pwr.ISO
Crew Switches Enable	Input.Crew.Out.Open.Rtn.ISO
Crew Switches Close	Input.Crew.Out.Close.ISO
Emergency Handle Sensing Switch	Input.DO.EmerHandleSwitch.ISO
Door Cut-Out Mechanical Lock Normal	Input.DPCutoutSwitchL.ISO
Panel Sensing Switch Left	Input.DO.PanelSSL.ISO
Panel Sensing Switch Right	Input.DO.PanelSSR.ISO
Lock Sensing Switch Left	Input.DO.SolenoidLockSSL.ISO
Lock Sensing Switch Right	Input.DO.SolenoidLockSSR.ISO
Door Location	Input.DoorID
Door Closing Visual Alarm	Output.PassAlertLight.ISO
Door Closing Audible Alarm	Output.PassAudibleSignal.ISO
Passenger Door Enable Light	-
Motor Current	Input.DO.MOTOR.CURRENT
Motor Encoder Phase A	Input.DO.MotorEncPhaseA.ISO
Motor Encoder Phase B	Input.DO.MotorEncPhaseB.ISO
Motor Encoder Index	Input.DO.MotorEncIndex.ISO
PWM	CPU.PWM.MOTOR
DCU Internal Signal	
Door Closing Visual Alarm Feedback	Input.PassAlertLight.Feedback
Door Closing Audible Alarm Feedback	Input.PassAudibleSignal.Feedback
Passenger Door Enable Light Feedback	-
CPU Open Interlock	CPU.OPEN_INTERLOCK.OUT
CPU Open Interlock Feedback	OPEN_INTERLOCK.FB
Normal Current Limit Lock	DO.CUR_LIMIT.NOR.ISO
High Current Limit Lock	DO.CUR_LIMIT.HI.ISO
Normal Current Limit Activation	CPU.CUR_LIMIT.NOR.ACT
Over-Current Shutdown	DCL.OV_CUR.SHD
Motor Voltage Reading Value	Input.DO.MOTOR.VOLT

Denver Door System Signal	Mnemonic in SRS
Motor Voltage Frequency	Input.MotorSupplyVoltage.FREQ
Logic Supply Monitoring	Input.VoltMonitLogicSupply
Encoder Supply Monitoring	Input.VoltMonitEncSupply
Motor Supply Monitoring	Input.VoltMonitMotorSupply
V-Gate Open Monitoring	VOLTMONIT.VGOPEN
V-Gate Close Monitoring	VOLTMONIT.VGCLOSE
Open Direction Motor	CPU.DIR.OPEN.MOTOR
Close Direction Motor	CPU.DIR.CLOSE.MOTOR
Motor Test	CPU.MOTOR.TEST
CPLD Data	DCL.DMS.DATA
CPLD Clock	CPU. DCL.CLOCK
CPLD Spare 1	CPU. DCL.SPARE1
CPLD Sare 2	DCL.CPU.SPARE2
Running LED	DMS.LED.Run
Warning LED	DMS.LED.Warn
Fault LED	DMS.LED.Fault
Test LED	DMS.LED.TEST
Spare Out	SPARE.OUT
Spare Out Feedback	SPARE.OUT.FB
Spare 1 Input	SPARE1.ISO
Spare 2 Input	SPARE2.ISO

3.1.2 External Interfaces

3.1.2.1 Specific Inputs and Outputs

Table 3-2 and Table 3-3 list the interfaces that are used in the Denver project only in addition to the generic signals.

Door Control Panel Interface

Table 3-2: List of the Input from the Door Control Panel

Name	Description	Source	Input / Output	Valid Range
Door Close Train Line	Pressing the door close pushbutton in the DCP generates pulse signal.	Door Control Panel	Input	0: Inactive 1: Active
Passenger Door Enable (+) Train Line	Pressing the passenger mode pushbutton in the DCP energizes this signal. The signal will be cleared when the door close pushbutton is pressed or the Zero Speed relay is off in the DCP.	Door Control Panel	Input	0: Inactive 1: Active
Passenger Door Enable (-) Train Line	Pressing the passenger mode pushbutton in the DCP energizes this signal. The signal will be cleared when the door close pushbutton is pressed or the Zero Speed relay is off in the DCP.	Door Control Panel	Input	0: Inactive 1: Active

Passenger Door Open Pushbutton (PDOP) Assembly Interface

Table 3-3: List of the Input and Output from the PDOP

Name	Description	Source	Input / Output	Valid Range
Passenger Door Enable Light	The DCU Application turns on this light when the Passenger Door Enable (+) and Passenger Door Enable (-) TL signals are active along with Zero Speed (+) and Zero Speed (-) CL signals. Otherwise the DCU Application will turn it off.	DCU	Output	0: Inactive 1: Active
Passenger Door Enable Light Feedback	This signal indicates whether the light is operating correctly.	Passenger Door Open Pushbutton	Input	0: Inactive 1: Active
Passenger Door Open Pushbutton	Pressing this pushbutton generates pulse signal to command doors to be open.	Passenger Door Open Pushbutton	Input	0: Inactive 1: Active

3.1.2.2 SPI Interface

The DCU Application communicates with the Neuron Gateway Application through the SPI interface. The protocol for this interface will be defined in the SPI Interface Control Document (ICD).

3.1.3 Specific Functions

This section specifies the function specific to the Denver project.

3.1.3.1 VAP5.1 Door Operator Movement Function

VAP5.1.7 Drive Door Open Sequence

VAP5.1.7.1 Valid Open Train Line Command:
VAP5.1.7.1.1 The DCU Application detects active Zero Speed (+) CL and Zero Speed (-) CL signals prior to Door Open (+) TL and Door Open (-) TL pulse signals.

VAP5.1.7.2 Valid Crew Switch Open Command:
VAP5.1.7.2.1 The DCU Application at location R1 and L1 detects active Zero Speed (+) CL and Zero Speed (-) CL signals prior to Crew Switch Open and Crew Switch Enable pulse signals.
VAP5.1.7.2.2 Crew Switch does not affect the Visual Alarm nor Audible Alarm

VAP5.1.7.3 Valid Passenger Door Open Pushbutton Command:
VAP5.1.7.3.1 The DCU Application detects active Zero Speed (+) CL and Zero Speed (-) CL signals prior to the Passenger Door Open Pushbutton pulse signal along with the active Passenger Door Enable (+) TL and Passenger Door Enable (-) TL signals.
VAP5.1.7.3.2 The presence of the “Zero Speed” Car Line signals, and the Passenger Door Enable” Train Line signals shall turn On the “Passenger Door Enable Light”

VAP5.1.1 Drive Door Close Sequence

VAP5.1.1.1 Loss of Zero Speed Car Lines:
VAP5.1.1.1.1 The DCU Application detects the loss of Zero Speed (+) CL or Zero Speed (-) CL signal.

VAP5.1.1.5 Valid Close Train Line Command:
VAP5.1.1.5.1 The DCU Application detects Door Close TL pulse signal.

VAP5.1.1.2 Valid Crew Switch Close Command:
VAP5.1.1.2.1 The DCU Application at location R1 and L1 detects Crew Switch Close pulse signal.
VAP5.1.1.2.2 Once the door close command is issued, the doors shall close without any delay before closing.
VAP5.1.1.3.2 Crew Switch does not affect the Visual Alarm nor Audible Alarm

VAP5.1.1.3 Valid Passenger Door Close Command:
VAP5.1.1.3.1 The DCU Application detects the expiration of the configurable delay (Passenger Door Open Duration in Table 3-4) when doors were opened by the Passenger Door Open command.

VAP5.1.1.4 Valid Loss of Passenger Door Enable Train Lines:
VAP5.1.1.4.1 The DCU Application detects the loss of Passenger Door Enable (+) TL or Passenger Door Enable (-) TL signal.

VAP5.1.3 Detect Obstruction

VAP5.1.3.1 Obstruction End Behavior:
VAP5.1.3.1.1 The DCU Application opens doors fully after the maximum number of the obstruction re-closing is over.

VAP5.1.3.2 Door Closing Visual Alarm and Door Closing Audible Alarm:
VAP5.1.3.2.1 The DCU Application lets the Door Closing Visual Alarm and Door Closing Audible Alarm turned on during the obstruction re-opening and re-closing VAP5.1.3.2.3 . The alarm and the light will go off when the doors are closed and locked VAP5.1.3.2.4 or the doors are fully open due to exceeding the maximum number of retries.

VAP5.1.4 Emergency Handle Switch Function

VAP5.1.4.1 Closing Door after an Emergency State – Previously in the Normal Mode:

The DCU Application resets the event “Emergency Handle Activation” and resumes to normal operation when:

VAP5.1.4.1.1 1) doors are closed and locked manually,
VAP5.1.4.1.2 2) the same-side active DCP gives valid close train line command,
VAP5.1.4.1.3 3) valid Crew Switch close command is issued at L1 and R1 location.

VAP5.1.4.2 Closing Door after an Emergency State – Previously in the Cut-Out Mode:
VAP5.1.4.2.1 The DCU Application resets the event “Emergency Handle Activation” and resumes to door cut-out mode when doors are manually closed and the Door Cut-out Mechanical Lock is in Cut-out position.

VAP5.1.6 Door Parameters

The Door Parameters are configurable with PTU Application with valid values seen in the Table 3-4.

Table 3-4: PTU Configurable Door Parameter

Full Name / Valid Values	Mnemonic in SRS
VAP5.1.6.1 **Door.Open**	
VAP5.1.6.1.1 Default opening time: 2s	OpeningTime
VAP5.1.6.1.2 Opening time range: 2.0 to 3.0s	OpeningTime.Range
VAP5.1.6.1.3 Opening time increment: 0.5s	OpeningTime.Increment
VAP5.1.6.2 **Door.Close**	
VAP5.1.6.2.3 Delay before Closing: 0 to 5s in 1s increment	ClosingDelay
VAP5.1.6.2.4 Default delay before closing time: 3s	-
VAP5.1.6.2.5 Default closing time: 2.5s	ClosingTime
VAP5.1.6.2.6 Closing time range: 2.0 to 3.0s	ClosingTime.Range
VAP5.1.6.2.7 Closing time increment: 0.5s	ClosingTime.Increment
VAP5.1.6.3 **Door.CrewSwitch**	
VAP5.1.6.3.2 Crew Switch Closing Time: Same as the “ClosingTime”	CrewSwitch.Outside.ClosingTime
VAP5.1.6.3.3 Crew Switch Opening Time: Same as the “OpeningTime”	CrewSwitch.Outside.OpeningTime
VAP5.1.6.13 **Door.Obstruction**	
VAP5.1.6.13.3 Distance of reopening after obstruction detection: 0, 3, 5, 10, 15, 20, and 25”	Obstruction.Distance
VAP5.1.6.13.4 Default distance of reopening after obstruction detection: 3”	-
VAP5.1.6.13.5 Obstruction retries limit: 0 to 5 attempts	Obstruction.Count
VAP5.1.6.13.6 Default obstruction retries: 3 attempts	-
VAP5.1.6.13.7 Default delay before re-closing: 2s	-
VAP5.1.6.13.8 Delay before re-closing: 0 to 5s in 1s increment	-
VAP5.1.6.5 **Passenger Door Open Duration**	
VAP5.1.6.5.2 Passenger Door Open Duration: 0 to 10mins in 1min increment	-
VAP5.1.6.5.3 Default passenger door open duration: 5mins	

3.1.3.2 Manage Snapshot Files

Table 3-5 describes the snapshot values that will be used for the Denver project.

Table 3-5: Snapshot Specification in Denver Project

Mnemonic in SRS	Full Name	Values
Door.Snapshot		
Snapshot.Period	Snapshot Period	50ms to 1s depending on a change in a signal
Snapshot.Capacity	Snapshot Capacity	41 Samples
Snapshot.SampleNbAfterevent	Snapshot Samples After event	20 Samples
Snapshot.NbOfFiles	Snapshot Number Of Files	20 Snapshot files

3.1.3.3 Signals and Function not Used in Denver Project

Some of the signals and functions described in the Generic DCM Application SRS are not used for this project.

Not used signals:

- Input.DO.MotorTemp
- Input.DO.EnclosureTemp.FREQ
- Input.IntCrewSwitch.HI.ISO
- Input.IntCrewSwitch.LO.ISO
- Input.Crew.In.Close.ISO
- Input.DPCutoutSwitchR.ISO
- Input.DeviceID
- Input.DO.EnclosureTemp
- Input.VoltMonitCommSupply
- DCL.HB_INTERLOCK.OUT
- DMS.LED.7SEG
- DMS.Display

Not used PTU configurable door parameter:

- OpeningDelay
- CrewSwitch.Outside.ClosingDelay
- CrewSwitch.Outside.OpeningDelay
- CrewSwitch.Inside.ClosingDelay
- CrewSwitch.Inside.ClosingTime
- CrewSwitch.Inside.OpeningDelay
- CrewSwitch.Inside.OpeningTime
- Obstruction.Range

Not used functions:

- SecondaryObstruction.Position
- SecondaryObstruction.Activation
- SecondaryObstruction.Delay
- Door.PartialOpen
- Door.DoorEnable
- Door.HalfOpen

3.1.4 VAP5.3 DCU Event List

This section list the event that is used for the Denver project.
There are three levels of the event depending on the severity of the condition. They are explained in the Section 1.3.1.2 at page 5.
Possible levels are:

- F: Failure
- E: Exception
- I: Information

There are two types of the event:

- P: Pulse
- S/R: Set / Reset

Table 3-6: Event List

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
1	F	No	No	No	No	2	0	No	VAP5.3.18 Watchdog Reset Detected	P	VAP5.3.18.1 The DCU logs this event when the processor has been reset by the watchdog timer circuit.
2	F	No	Yes	Yes	No	0	0	No	VAP5.3.19 DCU App. CRC32 Discrepancy	S/R	VAP5.3.19.1 The DCU sets this event when the CRC32 of the application does not match with CRC32 value stored in the Application table. VAP5.3.19.2 The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.
3	F	No	Yes	Yes	No	0	0	No	VAP5.3.20 Boot Loader CRC32 Discrepancy	S/R	VAP5.3.20.1 The DCU sets this event when the CRC32 of the Boot Loader application does not match with CRC32 value stored in the Boot Loader table. VAP5.3.20.2 The DCU resets this event when the CRC32 of the Boot Loader application matches with CRC32 value stored in the Boot Loader table.
4	F	No	Yes	Yes	No	0	0	No	VAP5.3.21 SRAM Failure	S/R	VAP5.3.21.1 The DCU sets this event when at least one SRAM cell or SRAM addressing does not work properly. VAP5.3.21.2 The DCU resets this event when the SRAM is fully functional.
5	F	No	Yes	Yes	No	0	0	No	VAP5.3.22 NVRAM Failure	S/R	VAP5.3.22.1 The DCU sets this event when at least one NVRAM cell or NVRAM addressing does not work properly. VAP5.3.22.2 The DCU resets this event when the NVRAM is fully functional.
6	E	No	No	No	No	0	0	No	VAP5.3.23 SPI Error	S/R	VAP5.3.23.1 The DCU sets this event when communication errors repeated on the Neuron SPI link. VAP5.3.23.2 The DCU resets this event when the communication is working without error.
7	E	No	No	No	Yes	0	0	No	VAP5.3.24 Motor Power not Available	S/R	VAP5.3.24.1 The DCU sets this event when the Motor Power input voltage is below functional range (<22 V). VAP5.3.24.2 The DCU resets this event when the Motor Power input voltage is within functional range (22-44 V).
8	I	No	No	No	Yes	3	3	No	VAP5.3.25 Motor Power Supply Below 30 V	S/R	VAP5.3.25.1 The DCU sets this event when the Motor Power input voltage is between 22 and 30 Volts. VAP5.3.25.2 The DCU resets this event when the Motor Power returns over 30 Volts.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
9	E	Yes	Yes	No	Yes	0	0	No	VAP5.3.26 Door Cut-Out Mechanical Lock in Not Normal	S/R	VAP5.3.26.1 The DCU sets this event when the Door Cut-Out Mechanical Lock is not in normal. VAP5.3.26.2 The DCU resets this event when the Door Cut-Out Mechanical Lock signal is in normal.
10	-	-	-	-	-	-	-	-	-	-	Unused.
11	E	No	No	No	Yes	0	0	No	VAP5.3.27 Emergency Handle Activation	S/R	VAP5.3.27.1 The DCU sets this event when the Emergency Handle signal is activated. VAP5.3.27.2 The DCU resets this event when the Emergency Handle signal is deactivated.
12	-	-	-	-	-	-	-	-	-	-	Unused.
13	E	Yes	Yes	No	No	0	0	No	VAP5.3.28 Door Open TLs Mismatch	S/R	VAP5.3.28.1 The DCU sets this event when there is a mismatch between the Door Open (+) TL and the Door Open (-) TL signals. VAP5.3.28.2 The DCU resets this event when both Door Open (+) and (-) TLs signals are active. ***VAP5.3.28.3 Note: The DCU does not set this event if the Door Cut-Out Mechanical Lock is not in Normal.***
14	E	Yes	No	No	No	1.4	0	No	VAP5.3.29 Door Unlocking Timeout	S/R	VAP5.3.29.1 The DCU sets this event when the Lock Sensing Switches indicate locked for more than 1.4s after an unlock command. VAP5.3.29.2 The DCU resets this event when the next time the Lock Sensing Switches indicate unlocked within 1.4s from the unlock command.
15	E	Yes	No	No	Yes	1	1	No	VAP5.3.30 Door Opening Time Longer than Expected	S/R	VAP5.3.30.1 The DCU sets this event when the door opening exceeds 1s more than the configured opening time. VAP5.3.30.2 The DCU resets this event when the door is fully open.
16	I	No	No	No	No	10	0	No	VAP5.3.31 Door Opening Timeout	S/R	VAP5.3.31.1 The DCU sets this event when the door opening exceeds 10s more than the configured opening time. VAP5.3.31.2 The DCU resets this event when the door is fully open.
17	E	Yes	No	No	Yes	0	0	Yes	VAP5.3.32 Loss of "Closed&Locked" Status w/o Door Command	S/R	VAP5.3.32.1 The DCU sets this event when doors are unlocked or open without an opening command or an emergency activation. VAP5.3.32.2 The DCU resets this event when doors are closed and locked.
18	E	Yes	No	No	No	1	1	No	VAP5.3.33 Door Closing Time Longer than Expected	S/R	VAP5.3.33.1 The DCU sets this event when the door closing exceeds 1s more than the configured closing time. VAP5.3.33.2 The DCU resets this event when doors are closed and locked.
19	E	Yes	No	No	No	1	1	No	VAP5.3.34 Door Locking Time Longer than Expected	S/R	VAP5.3.34.1 The DCU sets this event when it takes more than 1s from the configured closing time to lock doors. VAP5.3.34.2 The DCU resets this event when doors are closed and locked.
20	E	No	No	No	No	0	0	Yes	VAP5.3.35 Loss of Zero Speed with Door Open	S/R	VAP5.3.35.1 The DCU sets this event when any Zero Speed signal is lost with open or unlocked doors. VAP5.3.35.2 The DCU resets this event when both Zero Speed signals are present or doors are closed and locked.
21	E	No	No	No	No	0	0	No	VAP5.3.36 PSS Mismatch	S/R	VAP5.3.36.1 The DCU sets this event when there is a mismatch between left and right Panel Sensing Switch signals . VAP5.3.36.2 The DCU resets this event when both Panel Sensing Switch signals are consistent.
22	I	No	No	No	No	-	0	No	VAP5.3.37 Door Closing Obstruction	S/R	VAP5.3.37.1 The DCU sets this event when the DCU is unable to close doors after exceeding the number of configured obstruction retry. VAP5.3.37.2 The DCU resets this event when doors are closed and locked.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
23	F	Yes	Yes	No	No	2	0	No	VAP5.3.38 Motor Open Supply Error	S/R	VAP5.3.38.1 The DCU sets this event when the motor open supply signal is active without valid logic for more than 2s. VAP5.3.38.2 The DCU resets this event when the motor open supply signal is inactive.
24	F	Yes	Yes	No	No	0.2	0	No	VAP5.3.39 Solenoid Lock Supply Error	S/R	VAP5.3.39.1 The DCU sets this event when the solenoid lock supply signal is active without a valid logic for more than 0.2s. VAP5.3.39.2 The DCU resets this event when the solenoid lock supply signal is inactive.
25	-	-	-	-	-	-	-	-	-	-	Unused.
26	-	-	-	-	-	-	-	-	-	-	Unused.
27	E	Yes	No	No	Yes	0.1	0	No	VAP5.3.40 Motor Encoder Phase Lost Error	S/R	VAP5.3.40.1 The DCU sets this event when there are more than 20 transitions on one encoder phase and no transition on the other phase within 5ms. VAP5.3.40.2 The DCU resets this event whenever there are transitions on both phases.
28	F	Yes	Yes	No	Yes	0.1	0	No	VAP5.3.91 Motor Encoder Cable/Circuit Error	S/R	VAP5.3.91.1 The DCU sets this event when encoder phase A, phase B, and Index inputs are all on. VAP5.3.91.2 The DCU resets this event when any one of the encoder signal is off.
29	F	Yes	Yes	Yes	No	0	0	No	VAP5.3.42 Motor Control Module Communication Error	S/R	VAP5.3.42.1 The DCU sets this event when the motion control module is unable to handle requests from the door operator module. VAP5.3.42.2 The DCU resets this event when the next time a request is accepted.
30	E	Yes	No	No	No	0.2	0	No	VAP5.3.43 Motor Drive Error or Wire	S/R	VAP5.3.43.1 The DCU sets this event when the motor wire is disconnected or the motor does not respond for 0.2s. VAP5.3.43.2 The DCU resets this event when the motor responds to the DCU command within 0.2s.
31	F	No	No	No	No	0	0	No	VAP5.3.44 Motor Drive Circuit (H-Bridge) Error	S/R	VAP5.3.44.1 The DCU sets this event when the motor current flows with a single active transistor on the motor drive during the test. VAP5.3.44.2 The DCU resets this event when each transistor passes the test.
32	F	No	Yes	No	No	0	0	No	VAP5.3.45 Motor Current Reading Error	S/R	VAP5.3.45.1 The DCU sets this event when the motor current reading is out of the expected range. VAP5.3.45.2 The DCU resets this event when motor current reading is within range.
33	F	No	No	No	No	0	0	No	VAP5.3.46 Motor Voltage Reading Error	S/R	VAP5.3.46.1 The DCU sets this event when the motor voltage reading is out of expected range. VAP5.3.46.2 The DCU resets this event when motor voltage reading is within range.
34	F	Yes	Yes	No	No	0	0	No	VAP5.3.47 Normal Current Limit Error	S/R	VAP5.3.47.1 The DCU sets this event when the current limiting circuit does not cap the motor current at the normal current range. VAP5.3.47.2 The DCU resets this event when the current limiting circuit caps the motor current at the next door opening.
35	F	Yes	Yes	No	No	0	0	No	VAP5.3.48 High Current Limit Error	S/R	VAP5.3.48.1 The DCU sets this event when the current limiting circuit does not cap the motor current at the high current range. VAP5.3.48.2 The DCU resets this event when the current limiting circuit caps the motor current at the next door opening.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
36	E	Yes	Yes	No	No	0	0	No	VAP5.3.49 Current Level Selector Error	S/R	VAP5.3.49.1 The DCU sets this event when the DCU is unable to activate the normal current limit. VAP5.3.49.2 The DCU resets this event when the current limits test is executed successfully.
37	F	No	Yes	No	No	0	0	No	VAP5.3.50 Motor OC Shutdown Circuit Failure	S/R	VAP5.3.50.1 The DCU sets this event when the motor over-current shutdown circuit activation or deactivation time is out of tolerances when tested. VAP5.3.50.2 The DCU resets this event when the current limits test is executed successfully.
38	E	Yes	No	No	No	0	0	No	VAP5.3.51 Motor OC Shutdown Activated	S/R	VAP5.3.51.2 The DCU sets this event when motor over-current shutdown limit is reached, meaning that excessive current is used for 2s. VAP5.3.51.3 The DCU sets this event when the motor over-current shutdown circuit is inactive.
39	E	Yes	No	No	No	0	0	No	VAP5.3.52 5V Logic Supply Out of Range	S/R	VAP5.3.52.1 The DCU sets this event when the 5V Logic Supply is out of range. VAP5.3.52.2 The DCU resets this event when the 5V Logic Supply is within range.
40	E	Yes	No	No	No	0	0	No	VAP5.3.53 5V Encoder Supply Out of Range	S/R	VAP5.3.53.1 The DCU sets this event when the 5V Encoder Supply is out of range. VAP5.3.53.2 The DCU resets this event when the 5V Encoder Supply is within range.
41	E	Yes	No	No	No	0	0	No	VAP5.3.54 15V Motor Supply Out of Range	S/R	VAP5.3.54.1 The DCU sets this event when the 15V Motor Supply is out of range. VAP5.3.54.2 The DCU resets this event when the 15V Motor Supply is within range .
42	E	Yes	No	No	No	0	0	No	VAP5.3.55 V-Gate Open Supply Out of Range	S/R	VAP5.3.55.1 The DCU sets this event when the V-Gate Open Supply is out of range. VAP5.3.55.2 The DCU resets this event when the V-Gate Open Supply is within range.
43	E	Yes	No	No	No	0	0	No	VAP5.3.56 V-Gate Close Supply Out of Range	S/R	VAP5.3.56.1 The DCU sets this event when the V-Gate Close Supply is out of range. VAP5.3.56.2 The DCU resets this event when the V-Gate Close Supply is within range.
44	E	Yes	No	No	No	0	0	No	VAP5.3.57 Door Closing Visual Light OC	S/R	VAP5.3.57.1 The DCU sets this event when the feedback signal is inactive with a Door Closing Visual Light turned on. VAP5.3.57.2 The DCU resets this event when the feedback signal is active with a light turned on.
45	E	Yes	No	No	No	0	0	No	VAP5.3.58 Door Closing Visual Light On w/o Command	S/R	VAP5.3.58.1 The DCU sets this event when the feedback signal is active without a Door Closing Visual Light turned on. VAP5.3.58.2 The DCU resets this event when the feedback signal is inactive with a light turned off.
46	E	Yes	No	No	No	0	0	No	VAP5.3.59 Solenoid Low Side OC	S/R	VAP5.3.59.1 The DCU sets this event when the feedback signal is inactive with a Solenoid Low Side turned on. VAP5.3.59.2 The DCU resets this event when the feedback signal is active with a Solenoid Low Side turned on.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
47	F	Yes	Yes	No	No	0	0	No	VAP5.3.60 Solenoid Low Side On w/o Command	S/R	VAP5.3.60.1 The DCU sets this event when the feedback signal is active without a Solenoid Low Side turned on. VAP5.3.60.2 The DCU resets this event when the feedback signal is inactive with a Solenoid Low Side turned off.
48	E	No	No	No	No	0	0	No	VAP5.3.61 Spare Output OC	S/R	VAP5.3.61.1 The DCU sets this event when the feedback signal is inactive with a Spare Output turned on. VAP5.3.61.2 The DCU resets this event when the feedback signal is active with a Spare Output turned on.
49	E	No	No	No	No	0	0	No	VAP5.3.62 Spare Output On w/o Command	S/R	VAP5.3.62.1 The DCU sets this event when the feedback signal is active without a Spare Output turned on. VAP5.3.62.2 The DCU resets this event when the feedback signal is inactive with a Spare Output turned off.
50	F	No	No	No	No	0	0	No	VAP5.3.63 DCU Temperature Reading Error	S/R	VAP5.3.63.1 The DCU sets this event when the DCU Board Temperature reading is out of expected range. VAP5.3.63.2 The DCU resets this event when the reading is within range.
51	E	Yes	Yes	No	No	20	0	No	VAP5.3.64 Door Open Power TL Stuck	S/R	VAP5.3.64.1 The DCU sets this event when Door Open (+) TL signal is active for more than 20s. VAP5.3.64.2 The DCU resets this event when the Door Open (+) TL signal becomes inactive.
52	E	Yes	Yes	No	No	20	0	No	VAP5.3.65 Door Open Return TL Stuck	S/R	VAP5.3.65.1 The DCU sets this event when Door Open (-) TL signal is active for more than 20s. VAP5.3.65.2 The DCU resets this event when the Door Open (-) TL signal becomes inactive.
53	E	Yes	Yes	No	No	20	0	No	VAP5.3.66 Door Open Crew Key Stuck	S/R	VAP5.3.66.1.1 Monitored by L1 and VAP5.3.66.1.2 R1 DCU. VAP5.3.66.1 The DCU sets this event when the Door Open Crew Switch or the Crew Switch Enable signal is activated for more than 20 seconds. VAP5.3.66.2.1 Monitored by L1 and VAP5.3.66.2.2 R1 DCU . VAP5.3.66.2 The DCU resets this event when the stuck signal is deactivated.
54	I	No	No	No	No	0	0	No	VAP5.3.67 Event Log Erased by PTU	P	VAP5.3.67.1 The DCU logs this event when it receives a request to erase the event log from the PTU.
55	I	No	No	No	No	0	0	No	VAP5.3.68 Event Log Retrieved by PTU	P	VAP5.3.68.1 The DCU logs this event when it receives a request to get the event log from the PTU.
56	-	-	-	-	-	-	-	-	-	-	Unused.
57	I	No	No	No	No	10	0	No	VAP5.3.69 No Time for More Than 10 seconds	S/R	VAP5.3.69.1 The DCU sets this event when it has not received the present time after 10 seconds. VAP5.3.69.2 The DCU resets this event when it receives the present time.
58	I	No	No	No	No	0	0	No	VAP5.3.70 System Boot-Up	P	VAP5.3.70.1 The DCU logs this event after a DCU boot-up.
59	I	No	No	No	No	0	0	No	VAP5.3.71 Non-Volatile Data Corrupted	P	VAP5.3.71.1 The DCU logs this event when a structure in non-volatile memory has been reset to its default value due to CRC16 mismatch at boot-up.
60	-	-	-	-	-	-	-	-	-	-	Unused.
61	-	-	-	-	-	-	-	-	-	-	Unused.
62	F	No	No	No	No	0	0	No	VAP5.3.72 Domain Table Failure Detected by Neuron	S/R	VAP5.3.72.1 The DCU sets this event when the Neuron Firmware detects a failure. VAP5.3.72.2 The DCU resets this event when the Neuron Firmware error is cleared.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
63	F	No	No	No	No	0	0	No	VAP5.3.73 Neuron Application CRC32 Failure	S/R	VAP5.3.73.1 The DCU sets this event when the CRC32 of the Neuron Application does not match with CRC32 value stored in the Application table. VAP5.3.73.2 The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.
64	-	-	-	-	--		-	-	-	-	Unused.
65	E	Yes	No	Yes	No	20	0	No	VAP5.3.74 Door Close TL Stuck	S/R	VAP5.3.74.1 The DCU sets this event when the Door Close TL is active for more than 20s. VAP5.3.74.2 The DCU resets this event when the Door Close TL becomes inactive.
66	E	Yes	No	Yes	No	20	0	No	VAP5.3.75 Door Close Crew Switch Stuck	S/R	VAP5.3.75.1.1 Monitored by L1 and VAP5.3.75.1.2 R1 DCU. VAP5.3.75.1 The DCU sets this event when the Door Close Crew Switch signal is active for more than 20s. VAP5.3.75.2.1 Monitored by L1 and VAP5.3.75.2.2 R1 DCU. VAP5.3.75.2 The DCU resets this event when the Door Close Crew Switch signal becomes inactive.
67	E	No	Yes	No	No	0	0	No	VAP5.3.76 Crew Switch Signal at Invalid Location	S/R	VAP5.3.76.1 The DCU sets this event when any one of crew switch inputs is active at a non crew switch location. VAP5.3.76.2 The DCU resets this event when none of the crew switch inputs are active.
68	E	No	No	Yes	No	0	0	No	VAP5.3.77 LSS Mismatch When Unlocked	S/R	VAP5.3.77.1 The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with unlocked door. VAP5.3.77.2 The DCU resets this event when both Lock Sensing Switches signals are consistent with unlocked door.
69	E	No	Yes	No	No	0	0	Yes	VAP5.3.78 LSS Mismatch When Locked	S/R	VAP5.3.78.1 The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with locked door. VAP5.3.78.2 The DCU resets this event when both Lock Sensing Switches signals are consistent with locked door .
70	E	Yes	No	No	No	8	0	No	VAP5.3.79 Crew Switch Open Signals Mismatch	S/R	VAP5.3.79.1.1 Monitored by L1 and VAP5.3.79.1.2 R1 DCU. VAP5.3.79.1 The DCU sets this event when the Crew Switch Door Open signal is not consistent with Crew Switch Enable for 8s. VAP5.3.79.2.1 Monitored by L1 and VAP5.3.79.2.2 R1 DCU . VAP5.3.79.2 The DCU resets this event when Crew Switch Door Open and Crew Switch Enable are consistent.
71	E	Yes	No	No	No	3	0	No	VAP5.3.80 Door Summary Relay High Error	S/R	VAP5.3.80.1 The DCU sets this event when the DSR High is energized with open and unlocked door for 3s. VAP5.3.80.2 The DCU resets this event when signals are consistent. ***VAP5.3.80.3 Note: When DCML is not in normal, the DCU won't set this event.***
72	E	Yes	No	No	No	3	0	No	VAP5.3.81 Door Summary Relay Low Error	S/R	VAP5.3.81.1 The DCU sets this event when the DSR Low is energized with open and unlocked door for 3 seconds. VAP5.3.81.2 The DCU resets this event when the signals are consistent. ***VAP5.3.81.3 Note: When DCML is not in normal, the DCU won't set this event.***

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
73	E	Yes	Yes	No	No	2	0	Yes	VAP5.3.82 Zero Speed Carlines Mismatch	S/R	VAP5.3.82.1 The DCU sets this event when there is mismatch between (+) and (-) Zero Speed CL signals for 2s. VAP5.3.82.2 The DCU resets this event when both Zero Speed signals are consistent.
74	E	Yes	No	No	No	1	0	No	VAP5.3.83 Crew Switch w/o Zero Speed	S/R	VAP5.3.83.1.1 Monitored only by L1 and VAP5.3.83.1.2 R1 DCU. VAP5.3.83.1 The DCU sets this event when any one of Crew Switch signals is active without the presence of Zero Speed signals for 1s. VAP5.3.83.2.1 Monitored only by L1 and VAP5.3.83.2.2 R1 DCU. VAP5.3.83.2 The DCU resets this event when the Zero Speed signals are active or when the Crew Switch signals are inactive .
75	E	Yes	No	No	No	0	0	No	VAP5.3.84 Door Open PWR TL w/o Zero Speed	S/R	VAP5.3.84.2 The DCU sets this event when the Door Open (+) TL is activated without the presence of Zero Speed signals. VAP5.3.84.3 The DCU resets this event when the Zero Speed signals are active or when the Door Open (+) TL is inactive.
76	E	Yes	No	No	No	0	0	No	VAP5.3.85 Door Open RTN TL w/o Zero Speed	S/R	VAP5.3.85.1 The DCU sets this event when the Door Open (-) TL is activated without the presence of Zero Speed signals. VAP5.3.85.2 The DCU resets this event when the Zero Speed signals are active or when the Door Open (-) TL is inactive.
77	E	Yes	Yes	No	No	1	0	Yes	VAP5.3.86 Passenger Door Enable TL Mismatch	S/R	VAP5.3.86.1 The DCU sets this event when there is a mismatch signals between the (+) and (-) Passenger Door Enable TL signals. VAP5.3.86.2 The DCU resets this event when both Passenger Door Enable TL signals are same.
78	E	Yes	Yes	No	No	20	0	No	VAP5.3.87 Passenger Door Enable TL Stuck	S/R	VAP5.3.87.1 The DCU sets this event when the Passenger Door Enable TL signals are stuck high for more than 20s. VAP5.3.87.2 The DCU resets this event when both Passenger Door Enable TL signals are inactive.
79	E	Yes	No	No	No	0	0	No	VAP5.3.88 Passenger Door Enable TL w/o Zero Speed	S/R	VAP5.3.88.1 The DCU sets this event when either the Passenger Door Enable (+) TL or Enable (-) TL is activated without the presence of Zero Speed signals. VAP5.3.88.2 The DCU resets this event when both Passenger Door Enable TL signals are inactive or both Zero Speed signals are active.
80	E	Yes	No	No	No	20	0	No	VAP5.3.90 Passenger Door Open Pushbutton Stuck	S/R	VAP5.3.90.1 The DCU sets this event when the Passenger Door Open Pushbutton is stuck high for more than 20s. VAP5.3.90.2 The DCU resets this event when the Passenger Door Open Pushbutton is inactive.
81	E	Yes	No	No	No	0	0	No	VAP5.3.89 Passenger Door Enable Light OC	S/R	VAP5.3.89.1 The DCU sets this event when the feedback signal is inactive with a Passenger Door Enable Light turned on. VAP5.3.89.2 The DCU resets this event when the feedback signal is active with a light turned on.
82	E	Yes	No	No	No	0	0	No	VAP5.3.6 Passenger Door Enable Light On w/o Command	S/R	VAP5.3.6.1 The DCU sets this event when the feedback signal is active without a Passenger Door Enable Light turned on. VAP5.3.6.2 The DCU resets this event when the feedback signal is inactive with a light turned off.
83	E	Yes	No	No	No	0	0	No	VAP5.3.7 Door Open TL w/o Mechanical Lock Normal	S/R	VAP5.3.7.1 The DCU sets this event when the Door Open TL signal is activated without the Mechanical Lock in normal. VAP5.3.7.2 The DCU resets this event when the Door Open TL is inactive or when the Mechanical Lock is in normal.

Event #	Level	Snap	Dis. Open	Dis. Close	Red. Speed	Set T (s)	Reset T (s)	Initiate Close	Event Name	Type	Event Description
84	-	-	-	-	-	-	-	-	-	-	Unused.
85	F	No	Yes	No	No	0	0	No	VAP5.3.8 Invalid CPLD Version	S/R	VAP5.3.8.1 The DCU sets this event when the CPLD returns the invalid version or lower than the minimum expected version. VAP5.3.8.2 The DCU resets this event when the CPLD returns a version equals or higher than the minimum expected version.
86	E	No	No	No	No	0	0	No	VAP5.3.9 CPLD Test Version	S/R	VAP5.3.9.1 The DCU sets this event when the CPLD returns an unofficial version. VAP5.3.9.2 The DCU resets this event when the CPLD returns an official release version.
87	F	No	No	No	No	0	0	No	VAP5.3.10 Analog Reference Error	S/R	VAP5.3.10.1 The DCU sets this event when the analog reference value is more than 5% off. VAP5.3.10.2 The DCU resets this event when the analog reference vale is less than 3% off.
88	E	Yes	No	No	No	0	0	No	VAP5.3.11 Door Closing Audible Alarm OC		VAP5.3.11.1 The DCU sets this event when the feedback signal is inactive with a Door Closing Audible Alarm turned on. VAP5.3.11.2 The DCU resets this event when the feedback signal is active with an alarm turned on.
89	E	Yes	No	No	No	0	0	No	VAP5.3.12 Door Closing Audible On w/o Command		VAP5.3.12.1 The DCU sets this event when the feedback signal is active without a Door Closing Audible Alarm turned on. VAP5.3.12.2 The DCU resets this event when the feedback signal is inactive with an alarm turned off.
90	E	Yes	No	No	No	0	0	No	VAP5.3.13 No Open Interlock Feedback with Output On		VAP5.3.13.1 The DCU sets this event when the feedback signal is inactive with an Open Interlock turned on. VAP5.3.13.2 The DCU resets this event when the feedback signal is active with an Open Interlock turned on.
91	E	Yes	No	No	No	0	0	No	VAP5.3.14 Open Interlock Feedback On w/o Output On		VAP5.3.14.1 The DCU sets this event when the feedback signal is active without an Open Interlock turned on. VAP5.3.14.2 The DCU resets this event when the feedback signal is inactive with an Open Interlock turned off.
92	E	Yes	No	No	No	0	0	No	VAP5.3.15 Solenoid High Side OC		VAP5.3.15.1 The DCU sets this event when the feedback signal is inactive with a Solenoid High Side turned on. VAP5.3.15.2 The DCU resets this event when the feedback signal is active with a Solenoid High Side turned on .
93	E	Yes	Yes	No	No	0	0	No	VAP5.3.16 Solenoid High Side ON w/o Command		VAP5.3.16.4 The DCU sets this event when the feedback signal is active without a Solenoid High Side turned on. VAP5.3.16.3 The DCU resets this event when the feedback signal is inactive with a Solenoid High Side turned off.
94	E	Yes	Yes	No	Yes	0	0	No	VAP5.3.17 Motor Rotation with Door Locked		VAP5.3.17.1 The DCU sets this event when the motor is able to turn with door locked and closed. VAP5.3.17.2 The DCU resets this event when the motor does not turn with locked and closed door.

3.2 Software Design Description Document

3.2.1 Serial Driver Module SDD Document

The Generic Serial Driver Module SDD #201111582 has a complete design description for this project.

3.2.2 Watchdog Module SDD Document

The Generic Watchdog Module SDD #201111544 has a complete design description for this project.

3.2.3 Door Operator Module SDD Document

The Generic Door Operator Module SDD #201111568 requires an event code number for following events:

Table 3-7: Event Detected by Door Operator Module

Event Name	Event Code
Motor OC Shutdown Circuit Failure	37
Door Opening Time Longer than Expected	15
Door Closing Time Longer than Expected	18
Door Locking Time Longer than Expected	19
Normal Current Limit Error Detected	34
High Current Limit Error Detected	35
PSS Signal Error	21
Door Closing Obstruction Detected	22
Door Opening Timeout	16
Motor Control Module Communication Error	29
Current Level Selector Error	36
Motor Drive Circuit (H-Bridge) Error	31
Motor Rotation with Door Locked	94

3.2.4 Fault Logger Module SDD Document

The Generic Fault Logger Module SDD #201111583 has a complete design description for this project.

3.2.5 Proportional Integral Derivative Module SDD Document

The Generic Proportional Integral Derivative Module SDD #201111554 has a complete design description for this project.

3.2.6 Portable Test Application Module SDD Document

The Generic Portable Test Application Module SDD #201111586 requires an event code number for following events:

Table 3-8: Event Detected by the Portable Test Application Module

Event Name	Event Code
Event Log Erased by PTU	54
Event Log Retrieved by PTU	55

3.2.7 Real-Time Clock Module SDD Document

The Generic Real-Time Clock Module SDD #201111581 has a complete design description for this project.

3.2.8 Real Time Data Monitoring Module SDD Document

The Generic Real Time Data Monitoring Module SDD #201111585 has a complete design description for this project.

3.2.9 Neuron SPI Module SDD Document

The Generic Neuron SPI Module SDD #201111659 has a complete design description for this project.

3.2.10 Self-Test Module SDD Document

The Generic Self-Test Module SDD #201111584 requires an event code number for following events:

Table 3-9: Event Detected by Self-Test Module

Event Name	Event Code
DCU App. CRC32 Discrepancy	2
Boot Loader CRC32 Discrepancy	3
SRAM Failure	4
NVRAM Failure	5

4 PTU APPLICATION

4.1.1 Software Requirement Specification Document

The Generic Portable Test Application SRS #201111274 requires a specific requirement specification. There will be sub-sections in the next revision of this document.

4.1.2 Software Design Description Document

The Generic Portable Test Application SDD #201111295 requires a specific design description. There will be sub-sections in the next revision of this document.